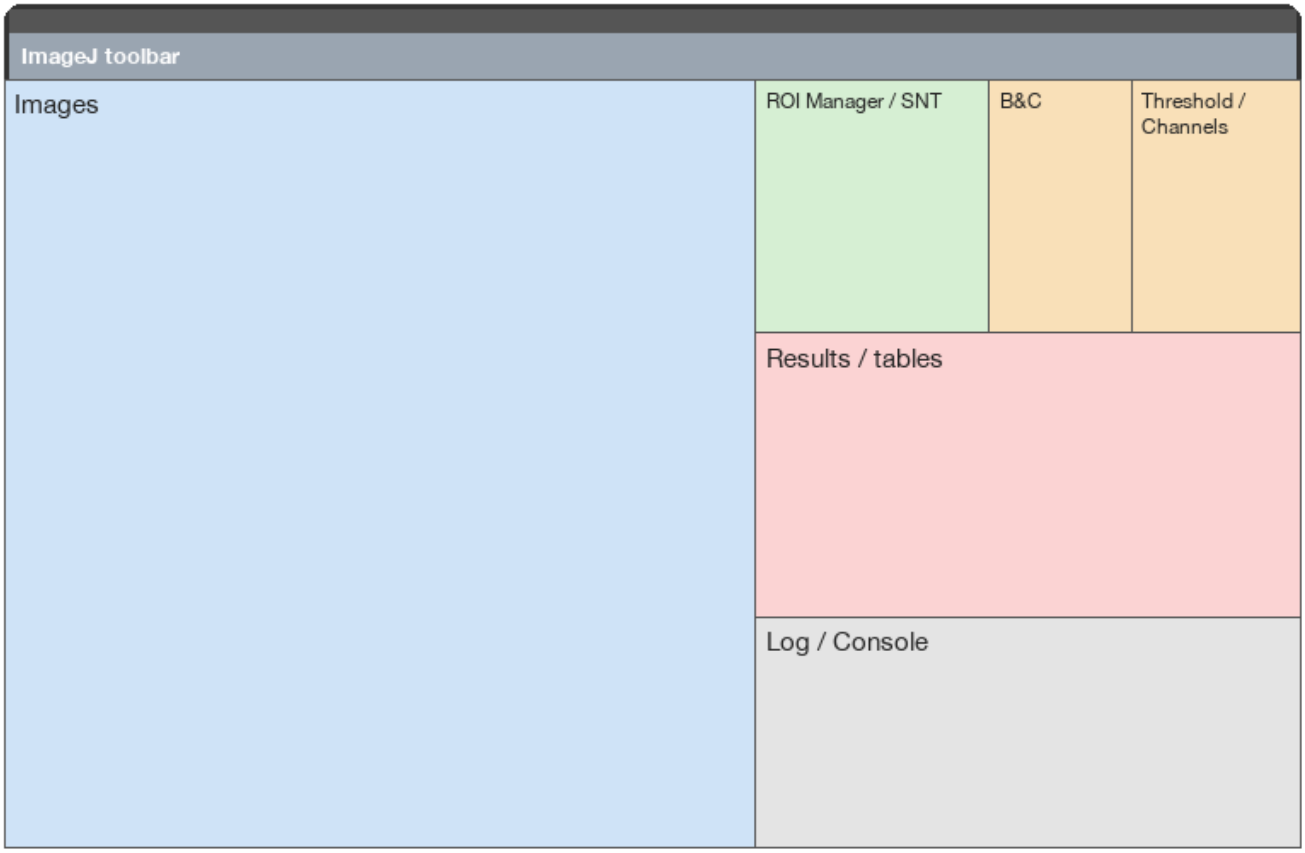
Window Layouts for Fiji

Fiji Window Layouts

Numbered window layouts, snap zones and exact saving of any arrangement, so that Fiji's windows are always where you want them.

Why · Highlights · Quick start · Installation · Built-in layouts · Menu reference



Layout 1

Why

Fiji opens every window wherever it likes, so a working session starts with dragging and resizing: images on top of each other, the ROI Manager behind an image, the Results table off screen. **Window Layouts** places Fiji's windows for you as they open, and puts everything back in order with one keystroke.

Highlights

Feature	What you get
Ten layouts, one key each	**Cmd**+**Option**+**0** to **9** on macOS, **Ctrl**+**Alt**+**0** to **9** on Windows and Linux, re-arranges every Fiji window at once
Automatic placement	images and tool windows go to their place in the active layout as soon as they open; images are tiled and zoomed so that the whole image shows (or, if you choose, so that it fills its area)
Six designs to start from	balanced, image first, analysis bench, side-by-side comparison, neuron tracing, two displays
Save exactly what you see	arrange the windows by hand and save the arrangement: every position, size and image zoom comes back
Snap zones	send the active window to a half, a quarter, a third or two thirds of the screen (top to bottom on a portrait display), or to another display
Trackpad navigation	two fingers pan a zoomed image in every direction; Ctrl + two fingers zoom smoothly around the pointer
Images that follow their window	drag a window's edge and the image scales to fill the window (or to fit in it); pan through the part that does not fit
Only Fiji's windows	other applications are never moved
Scriptable	every command works in macros, for example `run("Layout 3");`

Quick start

1. Install the two files (see Installation) and restart Fiji.
2. Open some images and tool windows (ROI Manager, B&C, Results): they go to their places in layout 1.
3. Press **Cmd**+**Option**+**2**, **3**, … (**Ctrl**+**Alt**+**2**, **3**, … on Windows and Linux) to try the other layouts.
4. Click an image and press **Cmd**+**Option**+← to snap it to the left half; **Cmd**+**Option**+**Delete** puts it back (on Windows and Linux: **Ctrl**+**Alt**+← and **Ctrl**+**Alt**+**Backspace**).
5. Arrange the windows the way you like and choose **Plugins ▸ Window Layouts ▸ Save Current Arrangement**… to keep it in any slot.
6. **Plugins ▸ Window Layouts ▸ Window Layouts Shortcuts** shows every shortcut inside Fiji.

On Windows and Linux

The shortcuts in this guide are written for macOS. On Windows and Linux the plugin uses **Ctrl**+**Alt** instead of **Cmd**+**Option**, by itself:

macOS	Windows and Linux
Cmd+**Option** + key	**Ctrl**+**Alt** + key
Cmd+**Option**+**Delete** (back to where it was)	**Ctrl**+**Alt**+**Backspace** (Ctrl+Alt+Delete belongs to the system)

- **AltGr keeps typing.** Windows and Linux report the AltGr key as Ctrl+Alt, and on many keyboards AltGr with a digit or a letter types a character (@, #, {, []). While a text field has the focus (a dialog, the macro editor, the Log), Ctrl+Alt+key is therefore left to the field. Click an image or a tool window to use the shortcuts.

- **If your system already uses Ctrl+Alt.** Some graphics drivers on Windows rotate the screen with Ctrl+Alt+arrows; Linux desktops use Ctrl+Alt with the arrows (workspaces), T (terminal) or L (lock). Such keys never reach Fiji. Choose another pair, for example **Alt+Shift**, in Plugins ▸ Window Layouts ▸ Window Layouts Preferences…; Window Layouts Shortcuts always lists the keys in use.

Three plugins that work together

Each plugin works on its own; installed together, they share the screen and each other's commands.

All three are written in Java 8 for any system Fiji runs on, and their shortcuts follow your system: **Cmd** on macOS, **Ctrl** on Windows and Linux. So far they have been tested on macOS only; reports from Windows and Linux are welcome in each repository's Issues.

Plugin	What it gives you
Side Toolbar	Fiji's toolbar as a vertical bar you can reshape freely, plus a shelf for open tools, live image previews and macro buttons
Window Layouts (this one)	Ten numbered window layouts, snap zones and exact saving of any arrangement, with a keyboard shortcut for each
Undo History	Multi-level undo and redo with a history panel: pixels, selections, overlays, display, calibration, ROI Manager and Results

- **Window Layouts** keeps every layout clear of Side Toolbar's bars and status strip, and re-arranges the windows when a bar changes size or place.
- **Side Toolbar**'s image previews offer Window Layouts' placements (halves, thirds, quarters, other display, layouts 0 to 9) and Undo History's steps in their right-click menu.
- **Side Toolbar** adds an Undo History button to the bar when Undo History is installed, and **Window Layouts** places the Undo History window like any other tool window.

The layouts in detail

- **Ten layout slots, 0 to 9, all editable.** Slots 1 to 6 start with six built-in designs; slots 0, 7, 8 and 9 start empty. You can save over any slot at any time, and **Reset Layout**… brings a built-in design back.
- **Automatic placement.** Images and tool windows go to their place in the active layout as soon as they open. Images are tiled and zoomed so that the whole image shows, or so that it fills its area when Fill the Window is chosen (see Trackpad, zoom and resizing).
- **A shortcut for each layout.** Press **Cmd+Option+0** to **9** at any moment to rearrange everything.
- **Snap zones, like Magnet, but only for Fiji.** Send the active Fiji window to a half, quarter or third of the screen, or to another display.
- **Compose your own layout.** Arrange the windows by hand and save the arrangement, or type the zones as numbers.

Everything is in the menu **Plugins ▸ Window Layouts**, and every setting is saved at once (it survives a crash or a forced quit).

Installation

The plugin consists of two files in your Fiji folder (the one that contains `plugins/` and `macros/`: `Fiji.app` on Windows and Linux; on macOS, `Fiji.app` or the folder that contains it):

- the plugin: `plugins/Fiji_Window_Layouts.jar`
- the startup hook, which starts it when Fiji opens (`run("Start Window Layouts");`): `macros/AutoRun/Window_L`

From a release. Download `Window_Layouts-<version>.zip` from the Releases page, unzip it, copy the two files to those places, and restart Fiji.

Without the AutoRun macro the menu commands still work, but the shortcuts and the automatic placement only start after you use one of the commands (or **Plugins ▸ Window Layouts ▸ Start Window Layouts**).

From the source (macOS; the script uses the Java inside Fiji), close Fiji and run:

```
git clone https://github.com/AIHector/Fiji-Window-Layouts.git
cd Fiji-Window-Layouts
./test.sh     # layout and zoom boundary/repaint checks
./build.sh    # compiles locally with Fiji's own Java; does not install
./build.sh --install  # installs both files; close Fiji first
```

`build.sh` takes the Fiji folder as an optional argument (`./build.sh /path/to/Fiji`).

To uninstall, delete the two files above.

How windows are grouped

A layout places kinds of windows. A window whose kind has no zone in the active layout, or that is not in this table, stays where it opens, with one exception: if it opens under the Side Toolbar's bars or its status strip (Fiji's Console, for example), it moves just beside them, keeping its size when it fits.

Kind (name used in zones)	Windows
`images`	every image window: single images, stacks, hyperstacks, composites, plots
b&c	Brightness/Contrast (B&C), Window/Level (W&L)
`roi`	ROI Manager
`threshold`	Threshold, Threshold Color
`channels`	Channels Tool
`results`	Results, Summary and every other ImageJ table or text window
`log`	Log
`console`	Fiji's Console (uses the `log` zone when a layout gives it none)
`snt`	SNT main window ("SNT v5.0.21"), Path Manager, Fill Manager, Neuroanatomy Shortcuts
(toolbar)	the ImageJ toolbar is put at the top of display 1 and stretched across its width (Preferences can turn the stretching off)

How each kind is placed:

- **Images** are tiled in a grid inside their zone, as square as possible (2 images side by side, 3 or 4 in 2×2, 5 or 6 in 3×2, …). Each image window takes its whole cell. By default (**Image zoom: fit**) the whole image is fitted inside it, as large as fits; when the image's shape differs from the cell, part of the window stays empty. With **Fill the Window** (chosen in the image's Zoom menu of the Side Toolbar, or in Preferences) the image is zoomed to cover the whole window, so there is no empty space; what does not fit is cut off at the right and the bottom, because the view always starts at the image's top-left corner (pan with two fingers, with the hand tool or by holding the space bar; ImageJ's small navigator box shows which part is in view). With **Fit and Shrink the Window** the whole image is fitted and the window then shrinks around it, so it is smaller than its cell but has no empty space. Preferences also offer **steps** (the whole image at ImageJ's standard zooms). **Plots** are redrawn at the size of their cell instead of being zoomed.
- **No space between windows.** Zones that touch share their edge to the pixel, the work area starts right under the toolbar, and every built-in layout tiles the whole screen.

- **Results, Log, Console and ROI Manager** are resized to fill their zone. ROI Manager is never made shorter than 220 px, so its buttons stay usable.
- **Fixed-size tools** (B&C, Threshold, Channels, SNT) keep their own size and are moved to the top-left corner of their zone.
- **Shared slots.** Some kinds share a slot, for example B&C, Threshold and Channels, since you rarely use all of them at once. If several are open together they **cascade** 24 px down-right, so every title bar stays visible.
- **No hidden title bars.** A fixed tool can be taller than its zone (B&C is 303 px). The window below it (Results, Log or ROI Manager) then gives up height from its top, so it starts just below the tool. If that would make it shorter than its minimum, it stays where it is.

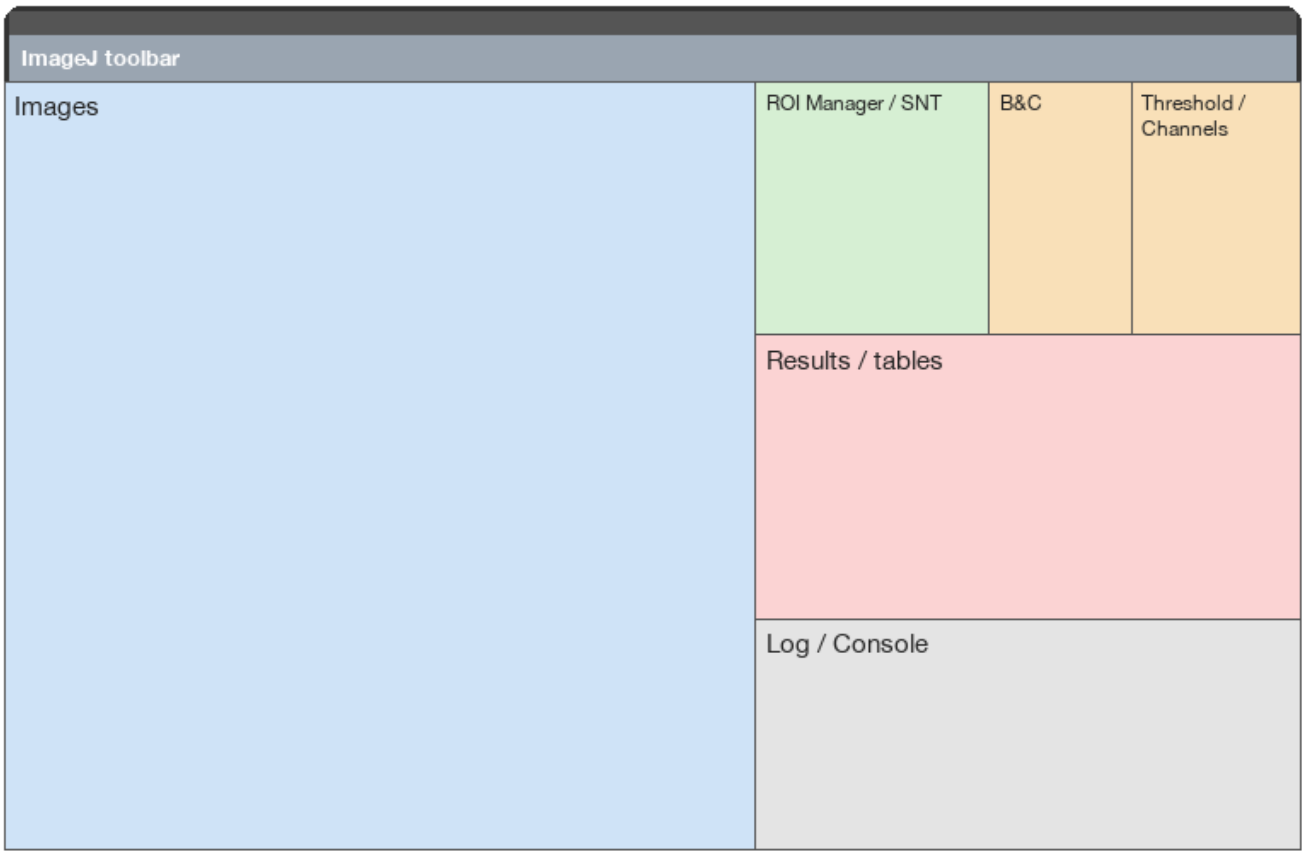
The built-in layouts

The built-in designs

The diagrams show your laptop screen (1512 × 982 points). On any screen the zones are percentages of the area below the ImageJ toolbar (the **work area**), so they scale. The tables give the zones in that unit, exactly as **Edit Layout**... shows them. You can copy a table into a custom layout and change it.

Layout 1: balanced

Images on the left 58%. On the right: ROI Manager, B&C and Threshold in a row, then Results, then Log.



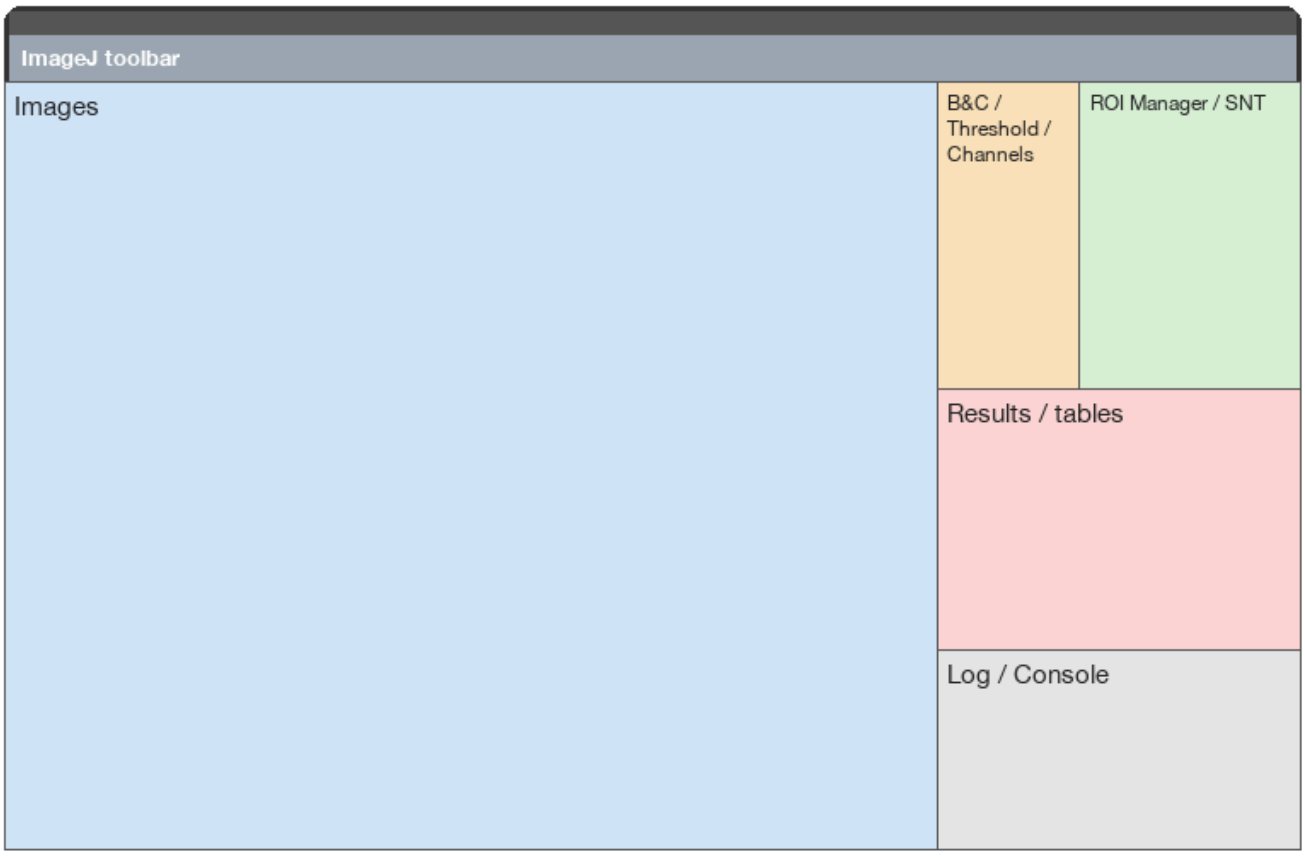
Layout 1

kind	zone x,y,w,h (% of work area)
images	0,0,58,100
b&c	76,0,11,33
roi, snt	58,0,18,33

kind	zone x,y,w,h (% of work area)
threshold, channels	87,0,13,33
results	58,33,42,37
log	58,70,42,30

Layout 2: image first

The largest image area (72%). One narrow column on the right: B&C next to the ROI Manager, then Results, then Log.

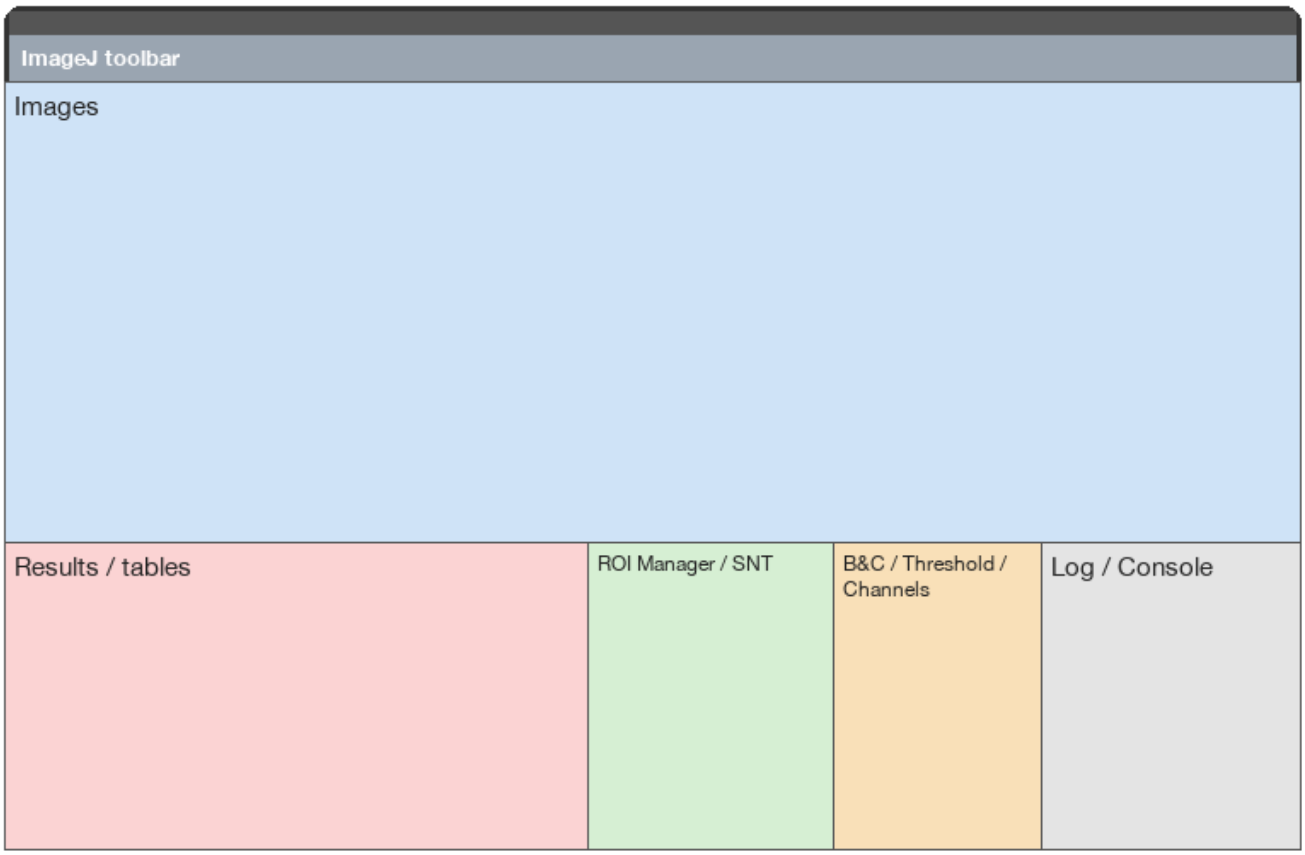


Layout 2

kind	zone x,y,w,h (% of work area)
images	0,0,72,100
b&c, threshold, channels	72,0,11,40
roi, snt	83,0,17,40
results	72,40,28,34
log	72,74,28,26

Layout 3: analysis bench

Images across the top 60%. A bench of tools along the bottom: Results, ROI Manager, B&C, Log.

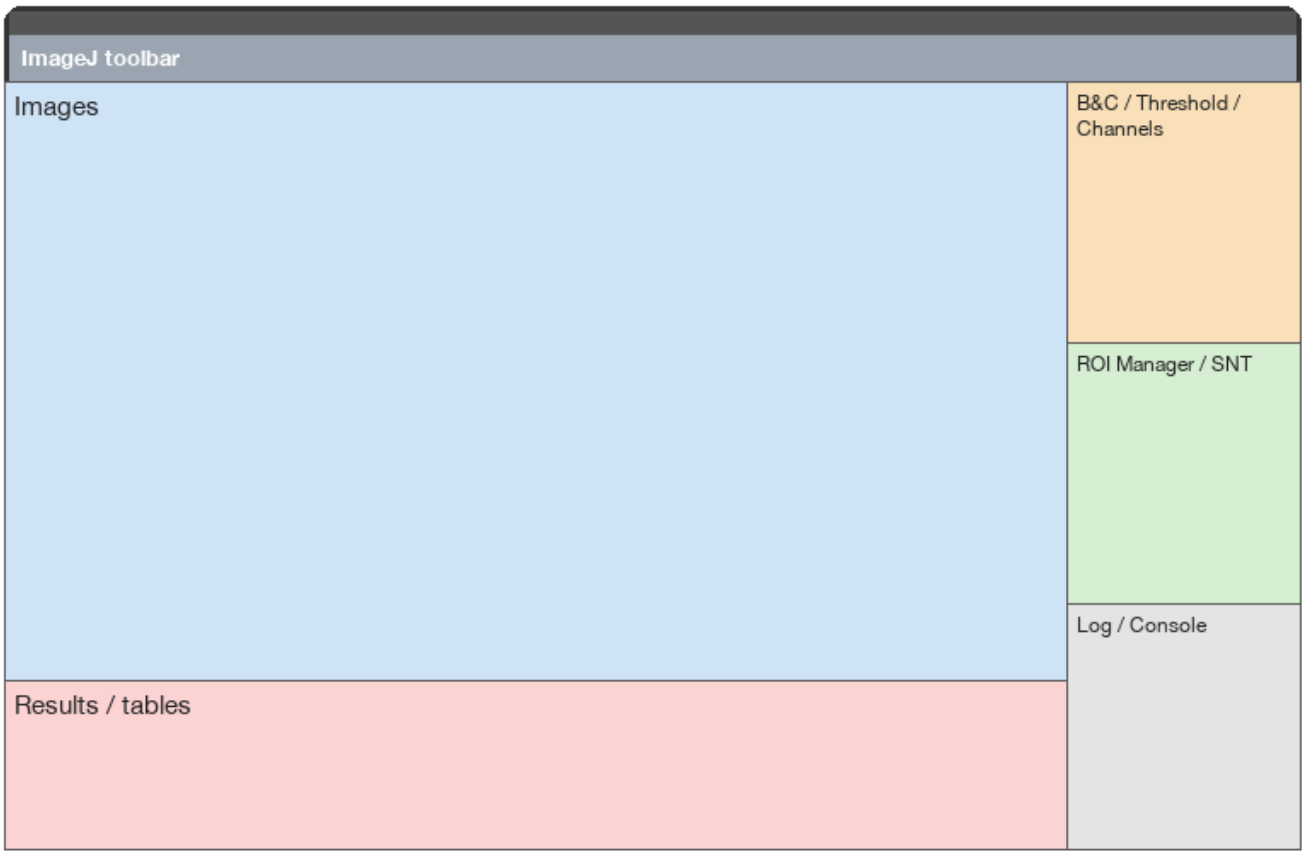


Layout 3

kind	zone `x,y,w,h` (% of work area)
images	`0,0,100,60`
b&c, threshold, channels	`64,60,16,40`
roi, snt	`45,60,19,40`
results	`0,60,45,40`
log	`80,60,20,40`

Layout 4: side-by-side compare

A wide image area for comparing images next to each other, a full-width Results strip below, tools in a column on the right.

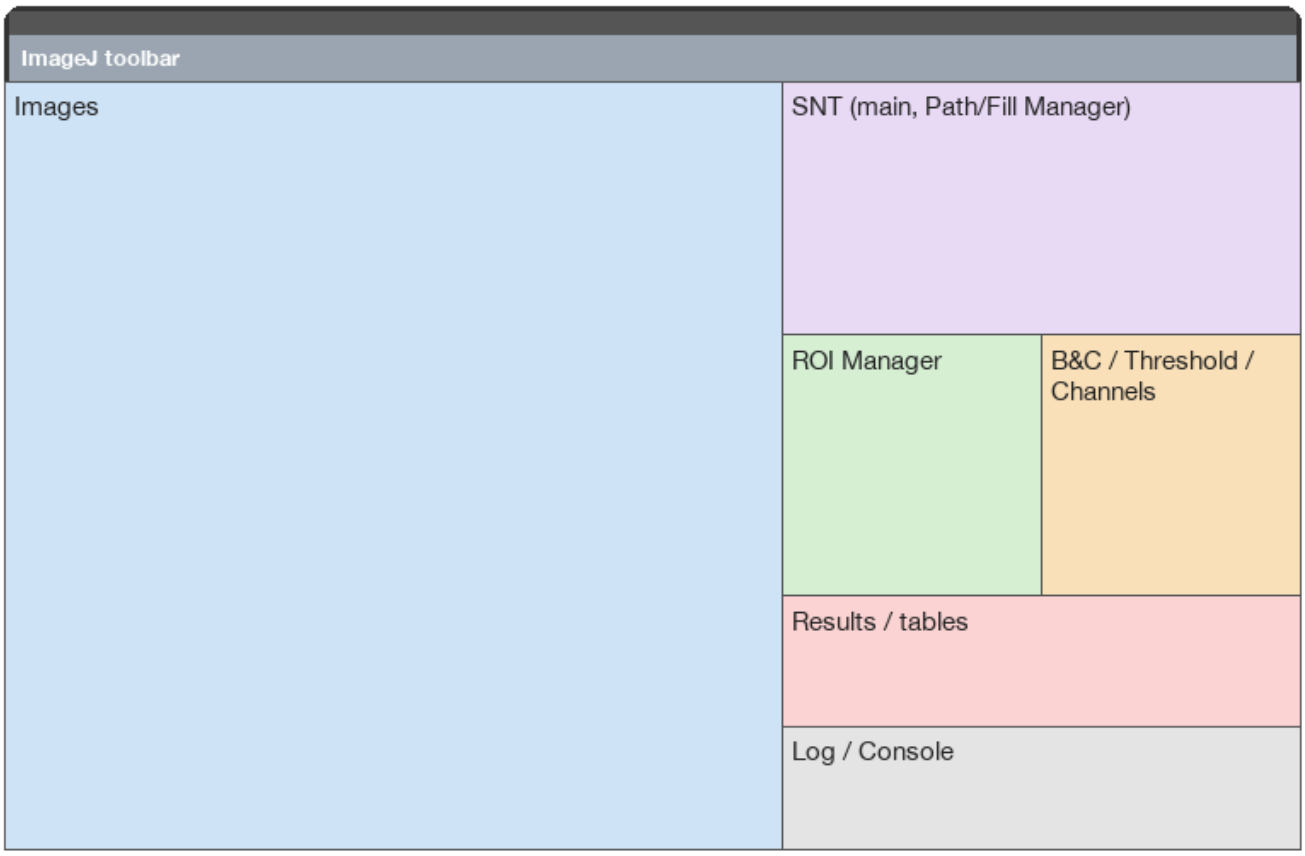


Layout 4

kind	zone x,y,w,h (% of work area)
images	0,0,82,78
b&c, threshold, channels	82,0,18,34
roi, snt	82,34,18,34
results	0,78,82,22
log	82,68,18,32

Layout 5: neuron tracing

Images on the left 60%. The SNT windows get the top right, with ROI Manager and B&C below, then Results and Log.



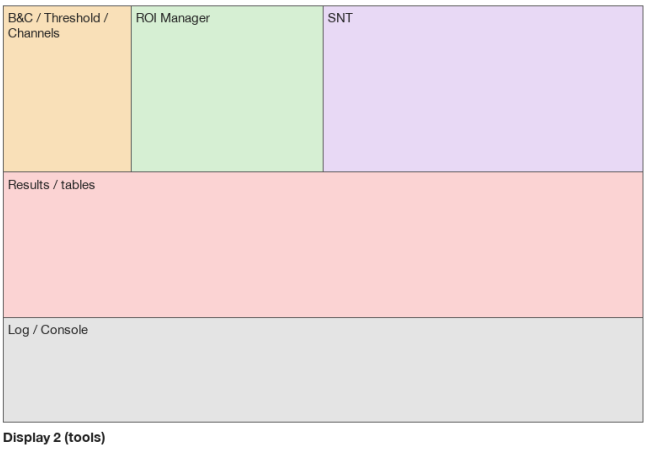
Layout 5

kind	zone `x,y,w,h` (% of work area)
images	`0,0,60,100`
b&c, threshold, channels	`80,33,20,34`
roi	`60,33,20,34`
results	`60,67,40,17`
log	`60,84,40,16`
snt	`60,0,40,33`

Layout 6: two displays

Images fill display 1 (the one with the toolbar); all tools go to display 2. **With only one display, layout 6 behaves like layout 2** and the status bar says so.





kind	zone x,y,w,h (% of work area)
images	0,0,100,100
b&c, threshold, channels	0,0,20,40@2
roi	20,0,30,40@2
results	0,40,100,35@2
log	0,75,100,25@2
snt	50,0,50,40@2

Your own layouts: every slot, 0 to 9

Save Current Arrangement

1. Arrange the windows by hand (or with the snap keys).
2. **Plugins ▸ Window Layouts ▸ Save Current Arrangement**…
3. Choose the slot (0 to 9) and type a name. The slot becomes the active layout. Saving into slots 1 to 6 replaces the built-in design until you reset it.

The plugin records **exactly what is on screen**:

- **Every image window**: its place, its size, its zoom and the part of the image in view.
- **Every other window, by its title**: B&C, ROI Manager, Results, Log, Console, Undo History, and any plugin's window.
- For each kind of window (images, B&C, ROI Manager…), also the area it covers, so windows of that kind that open later have a place too.

Press **Cmd**+**Option**+ the slot number to bring the arrangement back:

- Each window that was saved goes back to its exact place and size (to the pixel), on its display.
- An image that was open when you saved goes back to its own window, with its zoom. Other images take the remaining saved image windows, top-left first, zoomed to fit them as the Image zoom preference says. Images beyond the saved windows cascade from the last one.
- Windows saved by title are placed again when they open later, while that layout is active.

Positions are stored relative to the work area (the screen minus the Side Toolbar and the ImageJ strip), so a layout still fits if the bar changes width.

To set up layouts 1 to 4 for 1, 2, 3 and 4 images: open your tools and one image, arrange everything, save into slot 1; open a second image, arrange, save into slot 2; and so on. Re-arrange now (or a built-in layout) tiles the images first if you want a starting point: 2 images side by side, 3 images in three columns, 4 images in a 2 × 2 grid.

Edit Layout

Plugins ▸ Window Layouts ▸ Edit Layout… opens two dialogs:

1. **Edit slot**: which slot to write (0 to 9; default: the active one). **Start from**: which layout fills the second dialog (default: the active one). To start from a built-in layout and change it, choose it here.
2. One line per kind of window, plus a name.

Zone syntax:

You type	Meaning
`0,0,60,100`	x, y, width, height in **percent of the work area** (the screen below the ImageJ toolbar on display 1; the whole usable screen on other displays)
`12.5,0,30,40`	decimals use a **dot** (a comma separates the four numbers)
`0,50,100,50@2`	@2, @3… puts the zone on another display (see Multiple displays)
(blank)	this kind of window is not managed by the layout

Reset Layout

Plugins ▸ Window Layouts ▸ Reset Layout… removes what you saved in a slot: slots 1 to 6 return to their original design, slots 0, 7, 8 and 9 become empty.

The dialog refuses a zone that does not have four numbers, has a width or height of 0, or extends past 100% (for example `60,0,50,10`). It names the field and saves nothing. Typing `12,5` (a decimal comma) produces five numbers and is refused.

Empty slots show `(empty)` in the slot lists. Pressing the shortcut of an empty slot only shows "slot N is empty" in the status bar.

Keyboard shortcuts

All shortcuts use the **modifier**: **Cmd**+**Option** by default on macOS, **Ctrl**+**Alt** on Windows and Linux (see On Windows and Linux). It can be changed in Preferences. Magnet uses Ctrl+Option and Ctrl+Option+Cmd, so the two never collide. The shortcuts work while any Fiji window is in front (image, toolbar, Results, …), and they never reach other applications.

Modifier +	Action
0 … **9**	apply layout 0 … 9 (numeric keypad digits work too)
. or **=**	next layout (skips empty slots; 9 wraps to 0)
, or **-**	previous layout
A	turn automatic placement off / on (turning it on rearranges at once)
← / →	left / right half
↑ / ↓	top / bottom half
U / **I** / **J** / **K**	top-left / top-right / bottom-left / bottom-right quarter
S (or **D**) / **F** / **G**	left / center / right third (top / middle / bottom third on a display taller than wide)
E / **R** / **T**	left / center / right two-thirds (top / middle / bottom two-thirds on a display taller than wide)
C	center (keeps the size)
X	switch images between Fill the Window and Fit the Whole Image, applied at once to the active image
Return	maximize (the whole work area)

Modifier +	Action
Delete	restore the size and place from before the last snap
N or **]**	move the window to the next display
P or **[**	move the window to the previous display

Notes for this Mac:

- **Cmd+Option+D is taken by macOS** by default (show/hide the Dock), so it never reaches Fiji. Use **S** for the left third, or turn the Dock shortcut off in System Settings ▸ Keyboard ▸ Keyboard Shortcuts ▸ Launchpad & Dock.
- **Spanish keyboard.** Java reads keys by their position on a US keyboard. The keys labelled **.**, **,** and the digits work as printed, and so does the key labelled − (right of **.**). The **'** key right of **0** also works as previous layout. Keys that need Shift or Option on a Spanish keyboard (=, [,]) are matched by position: = is the ¡ key, [is the ` key,] is the + key. The letter alternatives (**.**, **„** **N**, **P**) are easier to remember.
- ImageJ's own single-key shortcuts are not affected: a key the plugin uses is taken before any window receives it, so ImageJ never acts on it too. (Up to 1.0.2, ImageJ also received the key and could reset the image's zoom right after a snap.) A key the plugin does not use, pressed with Cmd+Option, still reaches ImageJ. For example Cmd+Option+/ used to open Reslice, which is why / is now also previous layout.

Snap zones

Snapping acts on the **active Fiji window** (the one you clicked last). It never acts on the ImageJ toolbar. Zones are computed on the display the window is on, using the work area, so a snapped window never covers the toolbar.

- **Images**: the window takes the exact zone. By default the whole image is fitted inside it; with Fill the Window the image covers the whole zone, starting from its top-left corner; with Fit and Shrink the Window the whole image is fitted and the window shrinks around it (it starts at the zone's top-left corner).
- **On a display taller than wide**, thirds and two-thirds run from top to bottom: left = top, center = middle, right = bottom. Halves and quarters are unchanged.
- **Resizable windows** (Results, Log, ROI Manager, Console) take the exact zone.
- **Fixed-size windows** (B&C, Threshold…) are moved to the zone's corner and keep their size.
- **Restore** (Delete) returns the window to where it was before the last snap or display move.
- **Next / previous display** keeps the window's relative position, clamped so it stays on the target screen, and keeps an image's zoom. Displays are visited in the order 1 → 2 → 3 → 1.

The same zones are in the menu **Plugins ▸ Window Layouts ▸ Snap Zones** (halves, maximize, restore) and in **Snap Active Window**…, a dialog with all 19 zones. The Side Toolbar's Place on Screen menu lists them with their shortcuts.

Trackpad, zoom and resizing

These work in every image window (switch them in Preferences):

- **Two-finger panning.** On a zoomed image, two fingers on the trackpad move the image in every direction, like a map. This includes stacks and multichannel images: while the image is zoomed in, two fingers pan, and **Option** + **two fingers** change the slice or channel. When the whole image is visible (nothing to pan), up and down change the slice or channel as in ImageJ (as Edit ▸ Options ▸ Misc. decides).
- **Smooth zoom: Ctrl** + **two fingers.** Magnification and view position change continuously around the pointer. The view retains its fractional image-pixel position after scrolling stops: there is no timed handover, rounding step or finishing animation. The image canvas stays in its fitted area,

including during an unchanged-window layout. ImageJ's selection and overlay drawing uses the same fractional coordinate mapping. Zooming out stops at the scale that covers the current canvas. Genuine window resizing and explicit Fit/Fill commands still apply their requested new view. Shift + scroll keeps ImageJ's own zoom steps.

- **When an image window is resized** (dragging an edge or a corner), the image can:
  - **keep the zoom**: ImageJ's behaviour, more or less of the image shows (default);
  - **fill the window**: the image scales so that it covers the window, keeping the top-left corner of the visible part; for example a window as wide as the screen shows the image across the whole width, and two fingers pan through the part that does not fit. Placements (layouts and snap zones) then fill their zone too, from the image's top-left corner;
  - **fit the whole image**: the image scales so that all of it shows;
  - **fit and shrink the window**: the same when resizing; placements fit the whole image and shrink the window around it.

  The choice is in the Side Toolbar's image menu (Zoom ▸ When the window is resized) and in Preferences. When ImageJ itself zooms (+, −, the Zoom menu), it resizes the window around the new zoom; those resizes are left alone.
- **Right-click menu.** ImageJ's right-click menu on an image has a **Zoom** submenu: zoom in and out, 100%, original scale, to selection, 25%, 50%, 200% and 400% (the window keeps its size), Switch Fill / Fit, and the four ways images sit in their windows.
- **Pinch.** Java on macOS passes pinch gestures only to Swing windows, and ImageJ's image windows are AWT windows, so the plugin cannot receive a pinch. A window manager such as BetterTouchTool can send ImageJ's + and − keys for pinch gestures in Fiji only.

Automatic placement

When on (the default), new windows are placed according to the active layout as they open. When an image closes, the remaining images are re-tiled. A window you move by hand stays where you put it until the next window opens or closes (images re-tile) or you apply a layout.

- **Toggle Automatic Placement** (menu) or **Cmd+Option+A** turns it off and on. Turning it on applies the active layout immediately.
- **Re-arrange now** applies the active layout again, for example after moving windows by hand.
- If you prefer to arrange images by hand and only use the shortcuts, turn automatic placement off. Layout and snap shortcuts keep working.

Exact save: verification (22 October 2026)

In a separate copy of Fiji: three images (400×300, 256×256, 600×200) at chosen places and zooms, with B&C, Log and Undo History, saved into slot 7; after layout 2 rearranged everything, Layout 7 put every window back to the pixel (position and size) and gave each image its own zoom again (150%, 160%, 399%). Two new images then took the first free saved windows, top-left first. 536 unit tests pass, 17 of them for exact layouts.

With the Side Toolbar

When the Side Toolbar plugin shows Fiji's toolbar as a vertical bar, every layout fills the screen beside the bar, and below the ImageJ status strip (or above it, when the strip is at the bottom of the screen). The Side Toolbar places the ImageJ window itself, so Window Layouts no longer stretches it across the top. When the bar changes side or width, the windows are re-arranged.

Verified on 21 October 2026: layouts 1 to 5 with two images, ROI Manager, B&C and Log put every window beside the bar and below the strip; an image opened in the installed Fiji went to (40, 89), next to the 40 px bar and under the 55 px strip, and moved to (74, 89) when the bar grew to two columns. 519 unit tests (now 536), 4 of them for the bar's band.

Three displays, trackpad and resizing: verification (11 November 2026)

In the installed Fiji 2.18 on macOS with three displays (a laptop screen, a 2560 × 1440 monitor above it with the toolbar, a portrait monitor on the left), with real key presses and mouse drags:

- With Fill the Window, every snap shortcut on a 256 × 254 image gave a window of exactly its zone (for example halves 1260 × 1306, top and bottom halves 2520 × 653, thirds 840 × 1306, the whole work area 2520 × 1306), beside the Side Toolbar's bar and below its strip, with the image covering it from its top-left corner. In a separate copy of Fiji, with the default fit, each window again took its exact zone and showed the whole of a 1000 × 500 image (125% in a half, 83% in a third, 240% on the whole work area).
- Dragging the window's right edge to the screen's right side scaled the image with it (to 951%, filling the width); ImageJ's own + zoom was not undone.
- In a separate copy of Fiji (version 1.0.5): with Fit the Whole Image, a zoom with Ctrl + two fingers stayed where it was left (up to 1.0.4 this mode used ImageJ's Scale to Fit, which re-fitted the image after every zoom).
- In a separate copy of Fiji (version 1.0.4): with Fill the Window, zooming out stopped where the image still covered the window; Cmd+Option+X switched the image to Fit in the same window and back; zooming in and out in Fit kept the point under the pointer within 0.5 image pixels; the right-click menu ended with the Zoom submenu.
- In a separate copy of Fiji, on a 2000 × 2000 image with three channels (version 1.0.4): zoomed in, two fingers up and down panned without changing the channel; Option + two fingers changed the channel; with the whole image visible, up and down changed the channel.
- In a separate copy of Fiji: thirds and two-thirds on the portrait monitor ran from top to bottom (1440 × 843 and 1440 × 1686); smooth zoom over 200 steps kept the point under the pointer within 0.3 image pixels and returned exactly to the starting view; two-finger panning moved single images both ways and stacks sideways, while up and down changed the slice.

561 unit tests pass.

Multiple displays

Each window's display is found from where the window is, not from Java's own record, which can stay wrong after displays sleep, wake or reconnect (up to 1.0.2 this made snaps ignore the Side Toolbar's bars). Displays are numbered for `@N` and for display moves:

- **Display 1** is the display that holds the ImageJ toolbar (applying a layout puts the toolbar at its top-left).
- **Displays 2, 3,** … are the others, from left to right (then top to bottom).

For example, with the toolbar on a monitor above a laptop and a portrait monitor on the left: **1** = the monitor with the toolbar, **2** = the portrait monitor, **3** = the laptop screen. A zone on a display that is not connected falls back to display 1 (this rule is in the code but was not tried live).

Preferences

Plugins ▸ Window Layouts ▸ Window Layouts Preferences…

Option	Default	Effect
Shortcut modifier	`cmd+alt` (Cmd+Option)	also `ctrl+alt`, `ctrl+shift`, `alt+shift`. Ctrl+Option would clash with Magnet.
Place windows automatically as they open	on	same as Toggle Automatic Placement

Option	Default	Effect
Image zoom	fit	`fit`: whole image, as large as fits; `steps`: whole image at standard zooms. Fill the Window (below) makes images fill their zone instead
Images in their windows	keep the zoom	`keep the zoom`, `fill the window`, `fit the whole image`, `fit and shrink the window`: what placements and window resizing do with the image (see Trackpad, zoom and resizing)
Two-finger panning in images	on	pan a zoomed image with two fingers
Smooth zoom with Ctrl + two-finger scroll	on	off: Ctrl + scroll gives ImageJ's zoom steps
In stacks, two fingers pan a zoomed image	on	Option + two fingers then change the slice or channel; off: up and down always change the slice, as in ImageJ
Stretch the ImageJ toolbar across the screen	on	the toolbar takes the full width of display 1
Write a key log to `~/fijilayouts-debug.log`	off	records every Cmd/Option/Ctrl key the plugin sees, for troubleshooting shortcuts. Leave it off otherwise.

Menu reference (Plugins ▸ Window Layouts)

Command	What it does
Layout 0 … Layout 9	apply that slot (status message if it is empty)
Re-arrange now	apply the active layout again
Snap Active Window…	choose one of 18 snap zones for the active window
Save Current Arrangement…	store the current arrangement in any slot, 0 to 9
Edit Layout…	edit any slot numerically
Reset Layout…	slots 1–6 back to the original design, other slots emptied
Toggle Automatic Placement	on / off
Window Layouts Preferences…	modifier, automatic placement, key log
Window Layouts Shortcuts	the shortcut list, the active layout and the placement state
Snap Zones ▸ …	left / right / top / bottom half, maximize, restore
Start Window Layouts	starts the plugin (the AutoRun macro calls this)

All commands can be used in macros, for example `run("Layout 3");` or `run("Snap Active Window...", "zone=right-third");`.

Where settings are stored

In Fiji's preferences file, `~/Library/Preferences/IJ_Prefs.txt`, written immediately after every change:

Key	Meaning
`.fijilayouts.active`	active layout (0–9)
`.fijilayouts.auto`	automatic placement on/off
`.fijilayouts.mod`	shortcut modifier
`.fijilayouts.debug`	key log on/off
`.fijilayouts.custom.0` … `.9`	zones you saved in that slot (empty = built-in design or empty slot), e.g. `images=0,0,60,100;log=0,50,100,50@2`
`.fijilayouts.name.0` … `.9`	names of the saved layouts

To reset everything, quit Fiji and delete the lines starting with `.fijilayouts.`.

Troubleshooting

Problem	Fix
Shortcuts do nothing	Check that a Fiji window is in front. Check that `Window_Layouts_Startup.ijm` is in `Fiji/macros/AutoRun`, or run **Start Window Layouts**. Check the modifier in Preferences.
One shortcut does nothing	macOS or another app may own it (Cmd+Option+D = Dock). Turn on the key log in Preferences, press the key, and look at `~/fijilayouts-debug.log`: if the key is not listed, it never reached Fiji.
A window is not placed	Its kind is not in How windows are grouped, or the active layout leaves that kind blank, or automatic placement is off.
Tools overlap	Several fixed-size tools sharing one slot cascade over each other by design. On a small screen, close tools you do not need, or use layout 6 with a second display.
Part of an image is hidden	Fill the Window crops the edges so the window has no empty space. Pan with two fingers or the hand tool, or choose Fit the Whole Image or Keep the Zoom to see the whole image after a placement.
Fiji says “Plugin configuration error” at startup	A command name in `plugins.config` clashes with another plugin. Rebuild after renaming the entry.

Verification report (14–15 October 2026)

Everything below was tested on this Mac (Fiji 2.18.0 / ImageJ 1.54p, Java 21, macOS, three displays, Spanish keyboard). Positions were read from Fiji itself and cross-checked with macOS accessibility (System Events): both always agreed. Expected positions were recomputed independently from the zone percentages.

Passed

- 515 headless unit tests (`./test.sh`): zone parsing and validation, serialization round trips, built-in layouts (inside the screen, no overlapping zones), zone and snap arithmetic against hand-computed pixels, image grid, zoom choice, custom layout storage, next/previous skipping empty slots, key mapping, modifier matching, window classification, display order, layout 6 fallback rule.
- Fiji starts with no configuration error. The toolbar and the Log are placed at startup; the Console is placed when it opens.

- **Layouts 1–5** with 2 images + B&C + ROI Manager + Results + Log + Console: every window at its computed position, by automatic placement and by **real key presses** (Cmd+Option+1–5, physical key codes). 42/42 checks per layout, with no overlaps.
- **More windows**: 3 images (with a 20-slice stack), 4 images (with a 4000 × 3000 image, zoomed to 1/12), then a composite image, Channels, Threshold and a plot. Every window was at its expected position in all layouts; the plot was redrawn to its cell.
- **Layout 6** on the real second display (portrait monitor): images on display 1, all tools on display 2. 42/42.
- **All 18 snap zones** on an image, on Results (resizable) and on B&C (fixed size): 17/17 each, restore included. Halves, quarters, thirds, two-thirds, center and maximize were also tested with real key presses on an image.
- **Display moves** N, P,], [across all three displays, for an image, B&C and Results.
- **Next / previous layout** with real key presses (. , − =), including wrap-around past empty slots.
- **Re-tiling when an image closes**: 3 images → close one → the other two re-tiled and enlarged.
- **Toggle automatic placement**: menu and the toggle key (tested as Cmd+Option+0, then again as Cmd+Option+A after the change to ten slots). With it off, new windows are not moved; turning it on rearranges at once.
- **Empty slots**: status message, active layout unchanged.
- **Gapless layouts and your default** (15 October): slot 1 saved from your arrangement. After a restart, with an image, B&C, ROI Manager, Results, Log and Console open, every window was at its computed place and they tiled the screen with no gaps: toolbar 0–1512 × 34–120; image 0–892 × 120–934 (fill zoom 345%); ROI Manager 892–1358 and B&C 1358–1512 × 120–423; Log 423–701; Console 701–934. The Console, which Fiji re-shows rather than re-opens, is now placed too. 515 unit tests, including a check that every built-in layout tiles its work area exactly.
- **Ten editable slots** (15 October): Cmd+Option+0 on an empty slot 0 shows a status message. An arrangement saved into slot 0 came back with Cmd+Option+0 after switching to layout 2: every tool window returned exactly, and the images were re-tiled inside the saved image area. Built-in slot 3 was edited (images in the top half) and applied with Cmd+Option+3. Reset Layout then restored the original layout 3 (34/34 checks) and emptied slot 0. Next/previous layout cycle through 0–9. 505 unit tests.
- **Save Current Arrangement** → layout 1 → Cmd+Option+7: every window returned to its saved place (8/8).
- **Edit Layout**: both dialogs shown and confirmed through the real GUI (slot saved). Field values, blank fields, @2, and the error for an invalid zone (nothing saved) were tested through the same dialog in macro mode. The layout produced matched the typed percentages, including a Log zone on display 2.
- **Preferences**: dialog shown and confirmed. Switching the modifier (with Ctrl+Option chosen, Ctrl+Option+3 applied layout 3) and switching the key log off (no more lines written) were tested in macro mode.
- **Wrong modifiers ignored**: Ctrl+Option, Ctrl+Option+Cmd (Magnet's), Cmd+Option+Shift, Cmd alone, Option alone.
- **Shortcuts help** and **Snap Active Window** dialogs shown; the snap dialog moved the active window to the chosen zone.
- **SNT** (after the FlatLaf fix of 15 October): the main window "SNT v5.0.21" and Path Manager were recognised as SNT windows. With layout 5, a real Cmd+Option+5 put the main window at the top-left of the SNT zone (922, 126) and cascaded Path Manager 24 px down-right (946, 150). Both are taller than the zone, so they cover the ROI Manager / B&C area below; SNT windows keep their own size.
- **Persistence**: active layout, automatic placement, modifier and custom slots survive a File ▸ Quit and relaunch, and are on disk immediately after each change (before quitting).

Not verified

- **SNT Fill Manager and Neuroanatomy Shortcuts** windows (recognised by title, not opened live).
- **Your own fingers on the keyboard.** Key presses were sent as real macOS key events (System Events key codes) and as Java key events. Please try a few shortcuts yourself; if one fails, turn on the key log.

- **Layout 6 on a single display.** The fallback to layout 2 is unit-tested, but could not be tried live with three monitors connected.
- **The 3D Viewer, BigDataViewer and other special viewers.** They are treated as images but only moved, never zoomed; not tried.
- **Ctrl+Shift and Alt+Shift modifiers** were tested only in unit tests (Ctrl+Option was also tested live).
- **The Dock on the side or a different toolbar size.** The maths uses the real screen insets and toolbar height, but only this configuration was tried.

Known limitations

- On the 14-inch screen there is not enough room for every tool at once. B&C, Threshold and Channels cascade in one slot and can overlap a neighbouring window (Threshold is 303 px wide).
- A window moved to another display and back can end a few pixels from where it started (it is clamped to stay on screen). Delete (restore) returns it exactly.
- Snapping or rearranging re-zooms images (fit, steps or fill); it does not keep a zoom you set by hand.
- Fractional zoom is applied to standard ImageJ image canvases. Specialized viewers with their own canvas or layout keep their own zoom handling. Native image operations that read the source rectangle directly still see ImageJ's integer compatibility rectangle; standard mouse tools use the fractional coordinate-conversion methods. The adapter uses ImageJ internals for native overlay drawing and is verified with ImageJ 1.54p; other runtimes require validation.
- A resize that starts on a window edge covered by the Side Toolbar's bottom bar does not start: grab the edge above the bar.
- Fixed-size tools (B&C, Threshold, Channels) cannot be stretched: if their zone is larger than they are, some desktop shows around them.
- Custom layouts store one rectangle per kind of window, not per individual window.

Files

Path	Content
`src/fijilayouts/WindowLayouts_.java`	the plugin
`src/fijilayouts/Pan.java`	two-finger panning and smooth zoom
`src/fijilayouts/SmoothCanvas.java`	persistent fractional rendering, tool coordinates and native overlay drawing
`src/fijilayouts/Follow.java`	what an image does when its window is resized
`src/plugins.config`	its menu entries
`test/fijilayouts/LayoutTests.java`	headless unit tests (`./test.sh`)
`test/harness/fijilayouts/TestHarness.java`	development-only live-test helper (not installed)
`docs/make_diagrams.py`	draws the layout diagrams in this guide
`build.sh`, `test.sh`	build/install, unit tests
`README.md`, `README.pdf`	this guide
`LICENSE`	MIT License

License

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Version 1.0.8

Ctrl-scroll zoom retains its fractional view position throughout scrolling and after it stops, keeping the image steady under the pointer. The fitted canvas stays in place during unchanged-window layouts. Native mouse tools and overlays use the same view coordinates, and an obsolete placement retry cannot override a later gesture.

Verified with ImageJ 1.54p on macOS, including a real-trackpad acceptance check. Automated validation covers 561 existing layout checks, 24 zoom boundary/repaint checks, 27 image-window interaction checks and 3 checks with Undo History. Windows, Linux and other ImageJ runtimes remain unverified.

`./build.sh` builds locally without installing. `./test.sh` runs the existing layout and zoom boundary checks; `./test-integration.sh` checks wheel dispatch, fractional drawing, native tools, three-channel stacks and Fit/Fill interactions in a separate ImageJ with isolated preferences. `./test-combined.sh` checks the installed Undo History together with the new canvas. Use `./build.sh /path/to/Fiji --install` only with that Fiji closed.